# Amrutha varshini Anagurthi

**Data Engineer** | +1 3143350542 | amruthavarshinianagurthi@gmail.com

## Professional Summary

Data Engineer specializing in designing, building, and optimizing large-scale **Azure cloud data platforms**. Strong expertise in developing automated **ETL/ELT pipelines using Azure Data Factory, Azure Databricks (PySpark), and Airflow**, delivering up to **40% faster data processing and 99.5% pipeline reliability** for multi-terabyte datasets. Proven experience migrating **100TB+ of enterprise data to Azure Synapse Analytics and Snowflake**, architecting secure and cost-optimized **Azure Data Lake Storage Gen2** solutions, and driving **30% reductions in storage and compute costs**. Adept at enabling real-time analytics, governance, and scalable data architectures that translate directly into business outcomes.

## Technical Skills:

**Programming Languages:** Python (PySpark, Pandas, NumPy, Matplotlib), SQL (T-SQL, Azure Synapse SQL, Spark SQL), R (R Shiny), Scala

**Cloud Platforms & Services:** Microsoft Azure (Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Azure Data Lake Storage Gen2, Azure Blob Storage, Azure Event Hubs, Azure Stream Analytics, Azure Functions, Azure Logic Apps, Azure Service Bus, Azure SQL Database, Azure Cosmos DB, Azure Active Directory, Azure Key Vault, Azure Monitor, Log Analytics)

**Big Data Technologies:** Apache Spark (Spark Core, Spark SQL, Structured Streaming), Apache Kafka / Azure Event Hubs, Hadoop (HDFS, YARN), Hive, HBase

**Data Warehousing:** Azure Synapse Analytics, Snowflake, Oracle, Teradata

**Databases:** Azure SQL Database, Azure Cosmos DB, MySQL, PostgreSQL, MongoDB, SQL Server, Oracle, Azure Table Storage

**ETL / ELT Tools & Frameworks:** Azure Data Factory, Azure Databricks, Apache Spark, Custom Python & PySpark Pipelines

**Data Visualization & BI:** Power BI (Desktop, Service), Tableau, Azure Monitor Dashboards, R Shiny, Microsoft Excel

**DevOps & Infrastructure:** Azure DevOps CI/CD Pipelines, Bicep, ARM Templates,, Infrastructure as Code

**Data Formats & Storage:** Parquet, ORC, Avro, JSON, CSV, Azure Data Lake Storage Gen2, Azure Blob Storage, HDFS

**Development & Collaboration Tools:** Git, Azure Repos, Jupyter Notebooks, VS Code, PyCharm, Azure Data Studio, SQL Server Management Studio

**Data Modeling & Architecture:** Dimensional Modeling, OLTP/OLAP Design, Star,Snowflake Schemas, Data Normalization, Partitioning and Indexing Strategies

## PROFESSIONAL EXPERIENCE

### Cognizant Technology Solutions | Nov 2022 - July 2023 | Data Engineer

- Architected and developed scalable data pipelines using **Python and Apache Spark (PySpark, Spark Core, Spark SQL)** on **Azure Databricks and Azure Synapse Spark pools**, processing millions of records from diverse sources with modular design patterns and automated testing to ensure production reliability.
- Engineered end-to-end **ETL pipelines using Azure Data Factory and Azure Databricks**, ingesting and transforming financial data from **Azure Data Lake Storage Gen2 (Parquet, ORC)** into **Azure Synapse Analytics**, applying CI/CD practices, version control, and robust error handling.
- Built and optimized **real-time data streaming solutions using Azure Event Hubs and Apache Spark Structured Streaming**, enabling high-throughput ingestion with custom partitioning and data quality validations that improved processing efficiency by 30%.
- Designed and implemented observability and monitoring frameworks using **Azure Monitor, Log Analytics, and Application Insights**, enabling proactive detection of pipeline failures and data quality issues, reducing incident response time by 50%.
- Orchestrated complex data workflows using **Azure Data Factory pipelines and Azure Logic Apps**, integrating services such as **Azure Functions, Databricks, Synapse, Azure SQL, and REST APIs** to automate end-to-end data processing and improve operational reliability.
- Leveraged **Azure Databricks autoscaling clusters and Azure Synapse** for distributed Spark workloads, optimizing compute configurations and cost-efficient processing strategies to balance performance and infrastructure spend.
- Collaborated with product owners and business stakeholders to translate complex business requirements into technical epics and user stories, delivering scalable Azure-based data solutions that improved reporting accuracy and decision-making.
- Implemented **Infrastructure as Code (IaC)** using **Terraform and Azure Bicep**, automating provisioning of Azure resources including **Azure AD, ADLS Gen2, Azure Functions, Azure SQL, Service Bus**, ensuring consistent and repeatable deployments.
- Established data governance and security standards by configuring **Azure Active Directory RBAC, Managed Identities, and Key Vault**, enforcing fine-grained access controls and compliance with enterprise security requirements.
- Conducted rigorous code reviews and pair programming sessions, enforcing best practices for code quality, testability, and documentation across Azure-based data engineering projects.
- Developed and maintained detailed technical documentation covering Azure data architectures, transformation logic, and operational workflows to support maintainability and knowledge transfer.
- Applied **Agile/Scrum methodologies**, actively participating in sprint planning, standups, reviews, and retrospectives to ensure iterative delivery of high-quality Azure data solutions.
- Queried and analyzed large-scale datasets using **Spark SQL and Azure Synapse SQL**, delivering actionable insights that enhanced enterprise analytics and supported data-driven decisions.
- Automated data validation and quality checks by integrating **unit tests, integration tests, and data profiling** into **Azure DevOps CI/CD pipelines**, reducing production data defects by 40%.
- Led migration of enterprise data warehouses to **Azure Synapse Analytics**, designing optimized data loading strategies, partitioning schemes, and storage layouts that improved query performance and reduced costs.

### Cognizant Technology Solutions | Nov 2021 - Nov 2022 | Jr. Data Analyst

- Developed Python-based automation scripts for extracting and transforming large datasets (10M+ rows) from Azure data sources, reducing manual reporting effort by 40% and saving 10 hours per week.
- Built and published interactive **Power BI dashboards** with DAX measures, calculated columns, parameters, and optimized datasets, enabling real-time insights and faster business decision-making.
- Performed statistical analysis and data visualization using **Python (pandas, matplotlib)** and **R**, identifying trends and delivering actionable insights that improved business efficiency by 15%.
- Developed enterprise ETL workflows using **Informatica on Azure**, integrating **Azure SQL, Oracle, and Synapse**, ensuring data integrity and optimized performance across large-scale datasets.
- Implemented Spark DataFrame and Spark SQL solutions on **Azure Databricks** for processing complex Hive-compatible workloads, improving processing time by 30–35% and enabling near real-time analytics.
- Designed and optimized complex SQL queries using **Azure SQL and Synapse SQL**, leveraging joins, window functions, and performance tuning to reduce execution time by 25%.
- Designed normalized logical and physical data models for **Azure SQL–based OLTP systems**, improving data integrity, reducing redundancy, and enabling accurate reporting.
- Consolidated and maintained datasets across **Azure SQL, Synapse, Excel, and legacy systems**, creating unified reporting layers that streamlined analytics and operational reporting.
- Collaborated with data scientists to build advanced analytics and visualization solutions using **Power BI and R-based analytics**, supporting sophisticated analytical and forecasting use cases.
- Developed Python-based data integration utilities to consolidate data from multiple **Azure-hosted databases** into standardized formats for downstream analytics.
- Built data models and dashboards using **Azure Monitor and Log Analytics**, analyzing large-scale log data to provide real-time operational insights.
- Delivered recurring operational metrics, executive dashboards, and ad hoc reports through **Power BI and Azure Synapse**, improving visibility into KPIs and business performance.

## Education

**Master of Science Information Systems |** Saint Louis University, Saint Louis, MO | May 2025**.**

**Bachelors of Engineering in Computer Science |** Joginpally B.R Engineering college, Telangana, India | October 2021**.**